

★ **B.1-3**

Prove the generalization of equation (B.3), which is called the *principle of inclusion and exclusion*:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \\ & |A_1| + |A_2| + \dots + |A_n| \\ & - |A_1 \cap A_2| - |A_1 \cap A_3| - \dots \quad (\text{all pairs}) \\ & + |A_1 \cap A_2 \cap A_3| + \dots \quad (\text{all triples}) \\ & \vdots \\ & + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

B.1-4

Show that the set of odd natural numbers is countable.

B.1-5

Show that for any finite set S , the power set 2^S has $2^{|S|}$ elements (that is, there are $2^{|S|}$ distinct subsets of S).

B.1-6

Give an inductive definition for an n -tuple by extending the set-theoretic definition for an ordered pair.

B.2 Relations

A *binary relation* R on two sets A and B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we sometimes write $a R b$. When we say that R is a binary relation on a set A , we mean that R is a subset of $A \times A$. For example, the “less than” relation on the natural numbers is the set $\{(a, b) : a, b \in \mathbb{N} \text{ and } a < b\}$. An n -ary relation on sets $A_1, A_2, \dots, A_n$ is a subset of $A_1 \times A_2 \times \dots \times A_n$.

A binary relation $R \subseteq A \times A$ is *reflexive* if
 $a R a$

for all $a \in A$. For example, “=” and “ $\leq$ ” are reflexive relations on $\mathbb{N}$, but “<” is not. The relation R is *symmetric* if

$a R b$ implies $b R a$

for all $a, b \in A$. For example, “=” is symmetric, but “<” and “ $\leq$ ” are not. The relation R is *transitive* if

$a R b$ and $b R c$ imply $a R c$

for all $a, b, c \in A$. For example, the relations “<,” “ $\leq$,” and “=” are transitive, but the relation $R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1\}$ is not, since $3 R 4$ and $4 R 5$ do not imply $3 R 5$.

A relation that is reflexive, symmetric, and transitive is an *equivalence relation*. For example, “=” is an equivalence relation on the natural numbers, but “<” is not. If R is an equivalence relation on a set A , then for $a \in A$, the *equivalence class* of a is the set $[a] = \{b \in A : a R b\}$, that is, the set of all elements equivalent to a . For example, if we define $R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is an even number}\}$, then R is an equivalence relation, since $a + a$ is even (reflexive), $a + b$ is even implies $b + a$ is even (symmetric), and $a + b$ is even and $b + c$ is even imply $a + c$ is even (transitive). The equivalence class of 4 is $[4] = \{0, 2, 4, 6, \dots\}$, and the equivalence class of 3 is $[3] = \{1, 3, 5, 7, \dots\}$. A basic theorem of equivalence classes is the following.

Theorem B.1 (An equivalence relation is the same as a partition)

The equivalence classes of any equivalence relation R on a set A form a partition of A , and any partition of A determines an equivalence relation on A for which the sets in the partition are the equivalence classes.

Proof For the first part of the proof, we must show that the equivalence classes of R are nonempty, pairwise-disjoint sets whose union is A . Because R is reflexive, $a \in [a]$, and so the equivalence classes are nonempty. Moreover, since every element $a \in A$ belongs to the equivalence class $[a]$, the union of the equivalence classes is A . It remains to show that the equivalence classes are pairwise disjoint, that is, if two equivalence classes $[a]$ and $[b]$ have an element c in common,

then they are in fact the same set. Suppose that $a R c$ and $b R c$. Symmetry gives that $c R b$ and, by transitivity, $a R b$. Thus, we have $x R a$ for any arbitrary element $x \in [a]$ and, by transitivity, $x R b$, and thus $[a] \subseteq [b]$. Similarly, $[b] \subseteq [a]$, and thus $[a] = [b]$.

For the second part of the proof, let $A = \{A_i\}$ be a partition of A , and define $R = \{(a, b) : \text{there exists } i \text{ such that } a \in A_i \text{ and } b \in A_i\}$. We claim that R is an equivalence relation on A . Reflexivity holds, since $a \in A_i$ implies $a R a$. Symmetry holds, because if $a R b$, then a and b belong to the same set A_i , and hence $b R a$. If $a R b$ and $b R c$, then all three elements are in the same set A_i , and thus $a R c$ and transitivity holds. To see that the sets in the partition are the equivalence classes of R , observe that if $a \in A_i$, then $x \in [a]$ implies $x \in A_i$, and $x \in A_i$ implies $x \in [a]$.

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A binary relation R on a set A is *antisymmetric* if $a R b$ and $b R a$ imply $a = b$.

For example, the “ $\leq$ ” relation on the natural numbers is antisymmetric, since $a \leq b$ and $b \leq a$ imply $a = b$. A relation that is reflexive, antisymmetric, and transitive is a *partial order*, and we call a set on which a partial order is defined a *partially ordered set*. For example, the relation “is a descendant of” is a partial order on the set of all people (if we view individuals as being their own descendants).

In a partially ordered set A , there may be no single “maximum” element a such that $b R a$ for all $b \in A$. Instead, the set may contain several *maximal* elements a such that for no $b \in A$, where $b \neq a$, is it the case that $a R b$. For example, a collection of different-sized boxes may contain several maximal boxes that don’t fit inside any other box, yet it has no single “maximum” box into which any other box will fit.³

A relation R on a set A is a *total relation* if for all $a, b \in A$, we have $a R b$ or $b R a$ (or both), that is, if every pairing of elements of A is related by R . A partial order that is also a total relation is a *total order* or *linear order*. For example, the relation “ $\leq$ ” is a total order on the natural

numbers, but the “is a descendant of” relation is not a total order on the set of all people, since there are individuals neither of whom is descended from the other. A total relation that is transitive, but not necessarily either symmetric or antisymmetric, is a *total preorder*.

Exercises

B.2-1

Prove that the subset relation “ $\subseteq$ ” on all subsets of $\mathbb{Z}$ is a partial order but not a total order.

B.2-2

Show that for any positive integer n , the relation “equivalent modulo n ” is an equivalence relation on the integers. (We say that $a = b \pmod{n}$ if there exists an integer q such that $a - b = qn$.) Into what equivalence classes does this relation partition the integers?

B.2-3

Give examples of relations that are

- a.* reflexive and symmetric but not transitive,
- b.* reflexive and transitive but not symmetric,
- c.* symmetric and transitive but not reflexive.

B.2-4

Let S be a finite set, and let R be an equivalence relation on $S \times S$. Show that if in addition R is antisymmetric, then the equivalence classes of S with respect to R are singletons.

B.2-5

Professor Narcissus claims that if a relation R is symmetric and transitive, then it is also reflexive. He offers the following proof. By symmetry, $a R b$ implies $b R a$. Transitivity, therefore, implies $a R a$. Is the professor correct?

B.3 Functions